

Adaptive State-of-Health Forecasting for Second-Life LFP Cells Using Physics-Informed Synthetic Data and Hybrid Model Selection

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Second-life reuse of retired lithium iron phosphate (LFP) cells requires fast prediction of future degradation for each incoming cell. Instead of assuming a single degradation model is optimal for all cells, we develop an adaptive forecasting framework that selects between simple empirical extrapolation and a PyBaMM-trained neural operator based on early-cycle degradation complexity.

Approach (Figure 1). For each target cell the pipeline uses: (1) a short diagnostic characterisation together with the first $K=50$ operating cycles as early-cycle context; (2) a PyBaMM [1] SPMe corpus on Prada2013 LFP chemistry [2] with perturbed degradation trajectories around seven anchor cells from three suppliers (CALB, EVE, REPT), providing training trajectories that cover the second-life SoH range; (3) a Neural ODE [3] operator trained using PyBaMM-generated degradation trajectories and degradation-model-derived features, forecasting future SoH from the diagnostic features and the $K=50$ context window; (4) an exponential $SoH(n)=a+be^{-cn}$ baseline fitted on the same context window; and (5) a residual-based selector that predicts the last 20% of the context from a fit on the first 80% and picks the neural operator whenever the context-holdout residual exceeds a predeclared threshold τ , chosen on a synthetic-only selector pool disjoint from the external cells.

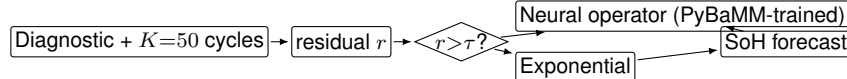


Figure 1: Adaptive selector: a context-holdout residual (exp fit on first 80% of the $K=50$ context, RMSE on last 20%) picks the neural operator above threshold τ ; τ is set on synthetic trajectories only.

Results. Across three external LFP cells excluded from operator training (CALB_0029, EVE_0003 and REPT_0028), exponential forecasting achieved 0.70 pp mean RMSE, while the PyBaMM-trained neural operator achieved 0.80 pp. The context-residual selector combined their complementary strengths, reducing mean RMSE to 0.47 pp and reproducing the oracle choice on all three evaluated cells: exponential forecasting for CALB_0029 and REPT_0028, and the neural operator for EVE_0003. These results motivate adaptive model selection rather than committing to either forecasting method alone.

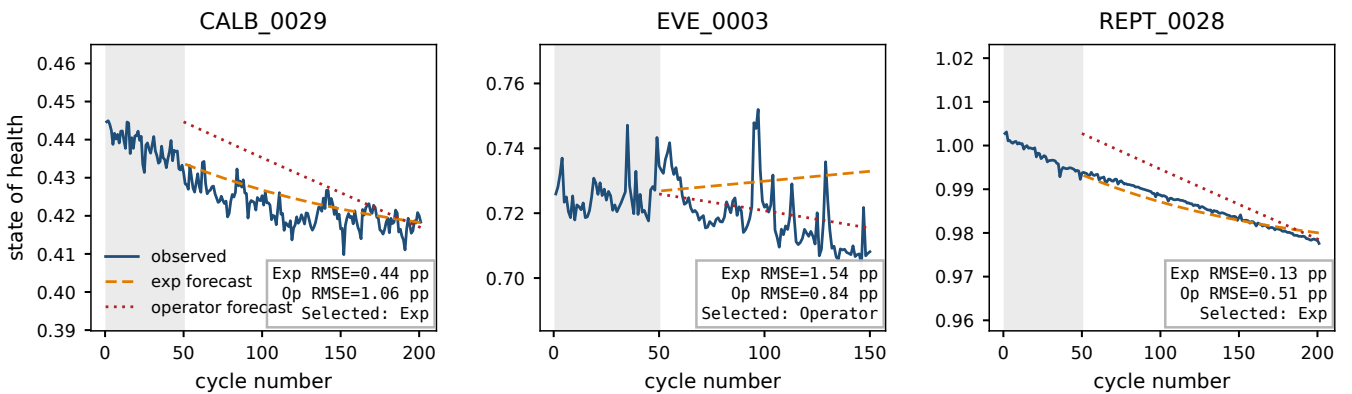


Figure 2: Observed and forecast SoH for the three external cells. Grey band = $K=50$ context; orange dashed = exponential forecast; red dotted = neural operator forecast. The context-holdout residual selects the exponential model for CALB_0029 and REPT_0028 and the operator for EVE_0003 – matching the per-cell oracle choice.

Ablations at matched split and seed indicate that PyBaMM-generated trajectories provide useful training diversity, while explicit degradation-parameter conditioning provides limited additional benefit (full-, no- and reduced- θ operators: 0.80, 0.82 and 0.90 pp mean).

The current evaluation is limited to three external cells, and longer-horizon projections require additional experimental validation.

[1] V. Sulzer et al., *J. Open Res. Software* **9** (2021), 14. [2] M. Prada et al., *J. Power Sources* **219** (2012), 353–362.

[3] R. T. Q. Chen et al., *NeurIPS* (2018).